RAG — Retrieval-Augmented Generation

What is RAG?

Retrieval-Augmented Generation (RAG) allows an AI model to answer questions using information retrieved from an external knowledge base. RAG first searches relevant company documents, retrieves useful passages, and gives them to the language model as context. The model then generates an answer based on that retrieved information. This is useful for private, organization-specific, or frequently changing information.

1. RAG Ingestion Diagram

Source Documents PDF • DOCX • FAQs	→	**Load & Clean** extract text	→	**Chunk Text** split passages	→	**Embeddings** vectors	→	**Vector DB** vectors + chunks + metadata

Ingestion: Documents are cleaned, divided into chunks, converted into embeddings, and stored in a vector database.

2. RAG Query Diagram

User Question asks a question	→	**Query Embedding** question → vector	→	**Retriever** similarity search	→	**Relevant Chunks** top matches	→	**LLM** context → answer

Query: The question is converted into a vector, relevant chunks are retrieved, and the LLM uses them to generate the answer.

Chunking and Why Chunk Size Matters

Chunking means dividing a document into smaller pieces before creating embeddings. Chunks should be large enough to preserve meaning but small enough to avoid unrelated information. Very small chunks can lose important context; very large chunks can reduce retrieval precision and use more model context. Chunk size also affects storage, retrieval speed, embedding cost, and answer quality. The best size depends on the document structure and the questions users ask.

3 Real Problems in a Petalnex-Type Business That RAG Can Solve

1. Internal knowledge search	Quickly find information in SOPs, policies, project documents, and technical files.
2. Customer-support knowledge	Retrieve relevant FAQs, product documents, policies, and previous solutions before answering customers.
3. Repeated operational questions	Answer recurring questions about workflows, onboarding, procedures, and service rules from the approved knowledge base.

Key idea: Ingestion prepares the knowledge base. Querying retrieves relevant knowledge and gives it to the LLM.